



**SILVER OAK
UNIVERSITY**
EDUCATION TO INNOVATION

**Aditya Silver oak Institute Of technology
Bachelor of Technology Department
of Information Technology (010)**

Caesar Cipher/Text Encryption

Group Members

Priya Purohit: 2502100330089

Vishva Purohit: 2502100330036

Riya Purohit: 2502100330037

Sejal Rajput: 2502100330018

Internal Faculty Guide

Prof. Zarna Shah

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Abstract

This project presents the Caesar Cipher, a simple yet innovative text encryption technique used for secure communication. It shifts each letter in the plaintext by a fixed number, converting readable text into coded form. The project applies design thinking by identifying the need for privacy, defining data security challenges, and creating a practical encryption solution. It demonstrates the basics of cryptography, encouraging creative and logical problem-solving for modern cybersecurity applications.

Introduction

The Caesar Cipher project demonstrates a simple yet effective text encryption method to secure digital communication. Each letter in a message is shifted by a fixed key, converting readable text into secret code. Using design thinking, the project addresses the need for data privacy, creating an easy-to-use encryption and decryption tool. It introduces basic cryptography concepts, promotes logical and innovative problem-solving, and lays the foundation for understanding modern digital security techniques.

Literature Survey/Research

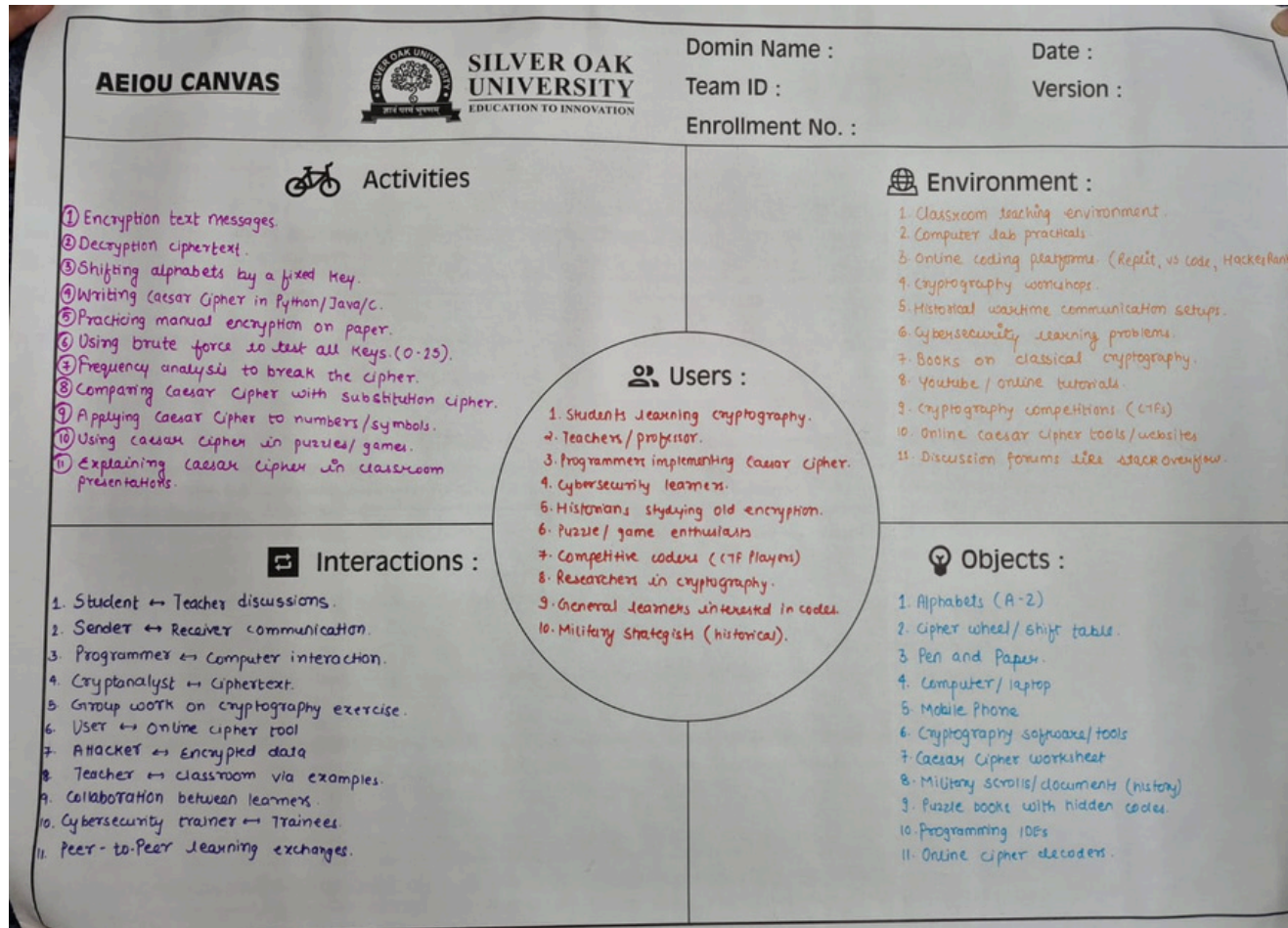
- The Caesar Cipher is one of the oldest encryption methods, used by Julius Caesar for secure communication.
- It works by shifting each letter in the text by a fixed key value.
- Forms the foundation of modern cryptography like AES and RSA.
- Widely used in learning, research, and basic data security demonstrations.
- Research highlights using design thinking to make encryption tools more user-friendly.
- Builds understanding of encryption, decryption, and data privacy principles.

Proposed Work

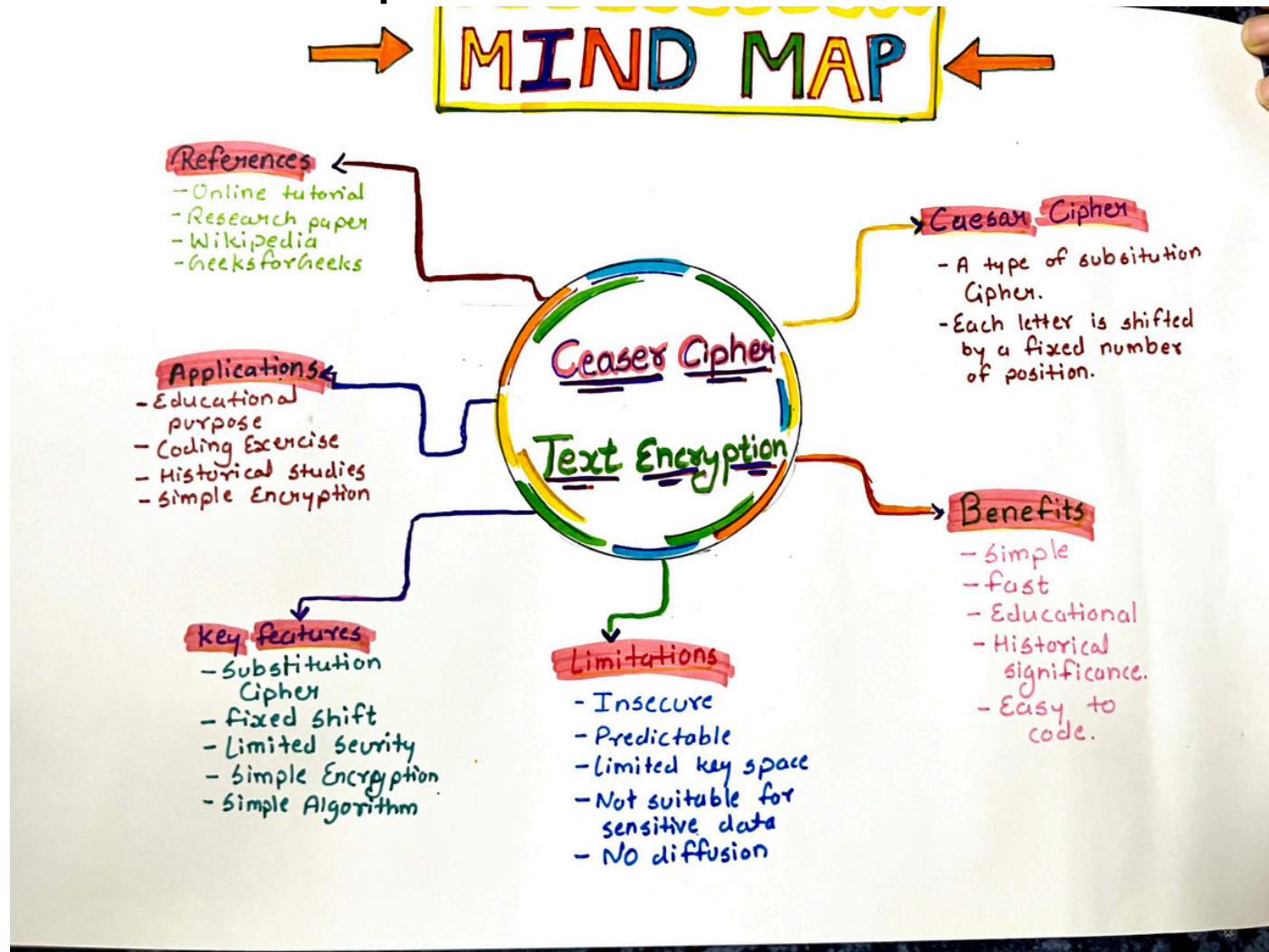
- Develop a simple encryption and decryption tool using the Caesar Cipher algorithm.
- The system will take plaintext and a key value as input and generate ciphertext by shifting letters.
- Implement the reverse process to decrypt the text and retrieve the original message.
- Use design thinking principles to make the tool easy, secure, and user-friendly.
- Conduct testing with different key values and text inputs to ensure accuracy.
- Focus on demonstrating data privacy, secure communication, and cryptographic logic.
- Present the project as a prototype for basic data security applications.

AEIOU Canvas

Diagram(UML)



Mind - Map



Ideation Canvas

IDEOATION CANVAS  SILVER OAK UNIVERSITY EDUCATION TO INNOVATION		Domain Name : Team ID : Enrollment No. :	Date : Version :
 Activities 1. Encrypting secret text messages. 2. Decrypting ciphertext back to plaintext. 3. Shifting alphabets with a fixed key. 4. Writing Caesar Cipher programs. 5. Solving cryptography puzzles using Caesar Cipher. 6. Demonstrating brute force attacks. 7. Frequency analysis on ciphertext. 8. Explaining encryption in workshops. 9. Comparing Caesar Cipher in games or escape rooms. 10. Comparing Caesar Cipher with modern ciphers. 11. Designing classroom activities with cipher wheels. 12. Applying Caesar Cipher in storytelling.	 People 1. Students learning cryptography. 2. Teachers and professors. 3. Programmers / software developers. 4. Cybersecurity learners. 5. Historians studying Caesar's military codes. 6. Puzzle / game enthusiasts. 7. Hackers / cryptanalysts. 8. CTF participants. 9. Military strategists. 10. General learners interested in codes. 11. Researchers in computer science. 12. Content creators making puzzle challenges.		
 Situation / Context / Location 1. Classroom teaching sessions. 2. Computer labs for practicals. 3. Cybersecurity workshops. 4. Historical military communication. 5. Coding competitions. 6. Puzzle / escape room games. 7. Online learning platforms. 8. Cryptography research seminars. 9. Interactive museum exhibitions. 10. Coding clubs / community hackathons. 11. Mobile learning apps. 12. Online cryptography forums / discussions.	 Props / Possible Solutions 1. Alphabet charts. 2. Cipher wheel / Shift table. 3. Pen and paper. 4. Computers / laptops. 5. Online Caesar Cipher tools. 6. Programming IDEs. 7. Whiteboard / markers for teaching. 8. Books / articles on ciphers. 9. Worksheets for practice. 10. Puzzle sheets / Crosswords with codes. 1. Create a Caesar Cipher learning app. 2. Design an interactive cipher wheel. 3. Build a puzzle / escape room game with Caesar cipher. 4. Implement a teaching kit for schools. 5. Make a website that auto-encrypts / decrypts Caesar cipher text. 6. Provide step-by-step coding tutorials. 7. Develop a quiz-based platform for cipher breaking. 8. Create a visual alphabet shift		

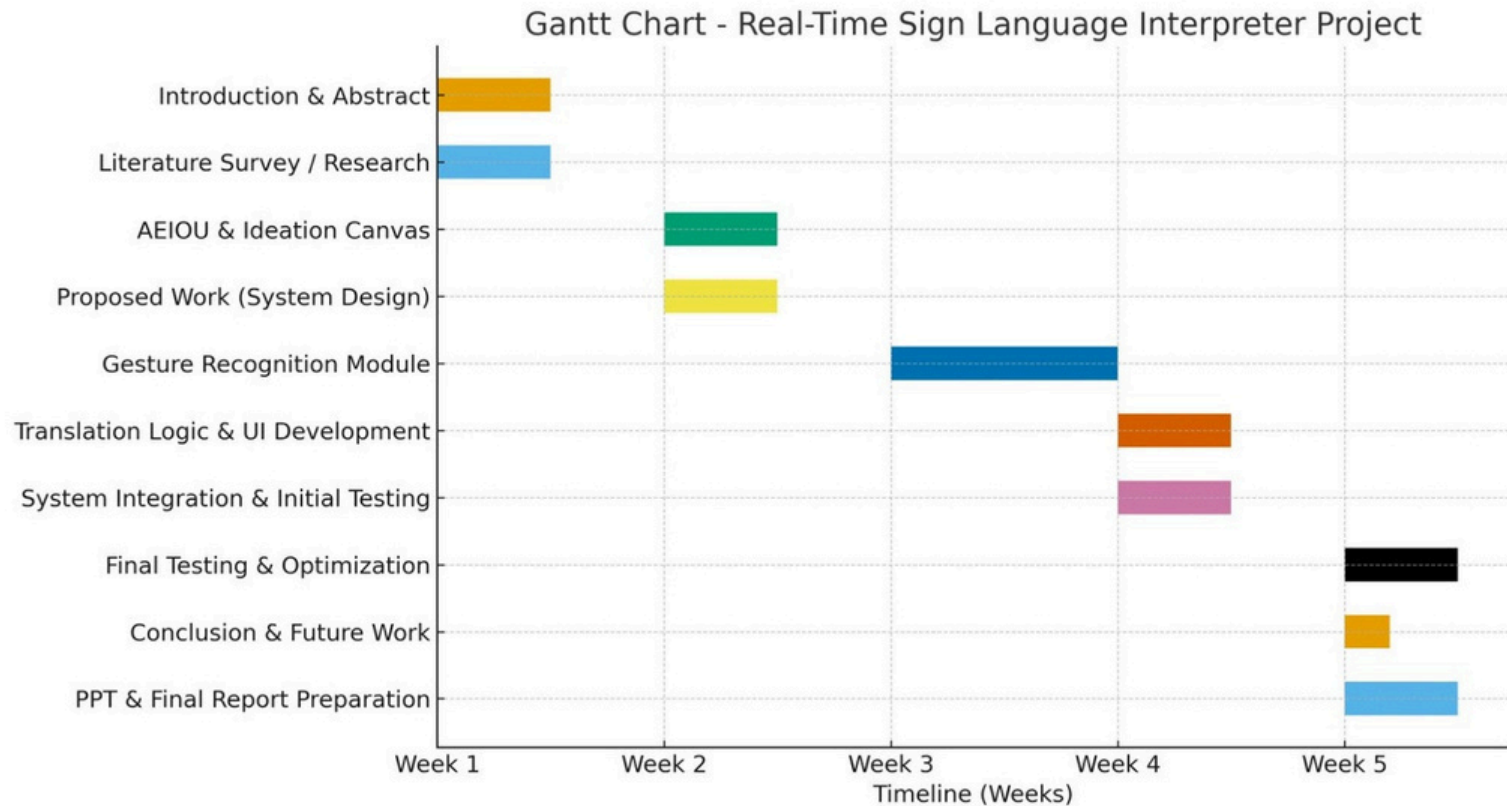
Product Development Canvas

 SILVER OAK UNIVERSITY EDUCATION TO INNOVATION		Domain Name : Team ID : Enrollment No. :	Date : Version :
PRODUCTION DEVELOPMENT CANVAS			
🔍 Problem Recognition → Many people want to send Confidential messages over the Internet. → Message can be intercepted by hackers or third parties. → Plain text Communications is not Secure. → Need a simple encryption method for data protection. → Lack of simple, educational tools to understand encryption concepts.	🛠️ Product Illustration (Prototype) → A simple software tool (desktop/web/mobile app). ⇒ User Interface Components : 1. Input box to enter plain text message. 2. Input box to set Key (Shift Value). 3. Button to Encrypt → Shows encrypted message. 4. Button to Decrypt → Converts encrypted message back to plain text. ⇒ Example Flow : → Enter message : "HELLO" → Enter key : 3 → Encrypt → "KHOOR" → Decrypt → "HELLO"		
💡 Problem Resolution ⇒ Algorithm Explanation : → Encryption : For each letter, shift by Key value (e.g., 3) in the alphabet. Formula : $C = (P + K) \bmod 26$ → Decryption : Reverse shift by key value to get original message. Formula : $P = (C - K + 26) \bmod 26$ ⇒ Benefits : → Provides a simple hands-on way to learn about cryptography. → Protects Sensitive messages from casual observers.	🔄 Product Remodelling (Redesign) ⇒ Improved Features : → Support for lowercase and uppercase letters. → Ability to handle Spaces and Special characters. → Friendly UI with instructions for beginners. → History log of encrypted/decrypted message. → Export/Save features for encrypted messages. ⇒ Future Enhancements : → Implement multiple encryption algorithms (e.g., Vigenère Cipher). → Allow random key generation. → Add Security level indicator.		

Future Work

- Enhance the Caesar Cipher by combining it with advanced algorithms like AES or RSA for stronger encryption.
- Develop a graphical user interface (GUI) for better usability and visual appeal.
- Implement encryption for numbers, symbols, and files, not just alphabets.
- Extend the project into a web or mobile application for real-world secure communication.
- Explore multi-level or hybrid encryption models to improve data protection.

Work carried out till date



Conclusion

The Caesar Cipher project successfully demonstrates how simple algorithms can ensure basic data security.

By applying design thinking, the project identifies the need for privacy and offers an innovative yet easy encryption solution.

It provides a strong foundation for understanding cryptography concepts, encouraging creative thinking and further research in digital security and information protection.

References

- William Stallings, Cryptography and Network Security: Principles and Practice, Pearson Education, 7th Edition, 2017.
- Behrouz A. Forouzan, Cryptography and Network Security, McGraw-Hill Education, 2015.
- Research Paper: Sharma, P. & Kumar, R. (2020). Study of Classical Encryption Techniques in Modern Cryptography. International Journal of Computer Applications.
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Thank You..!

Any Questions?